CSA1616 - DATA WAREHOUSING AND DATA MINING

Lab Experiments (1-3)

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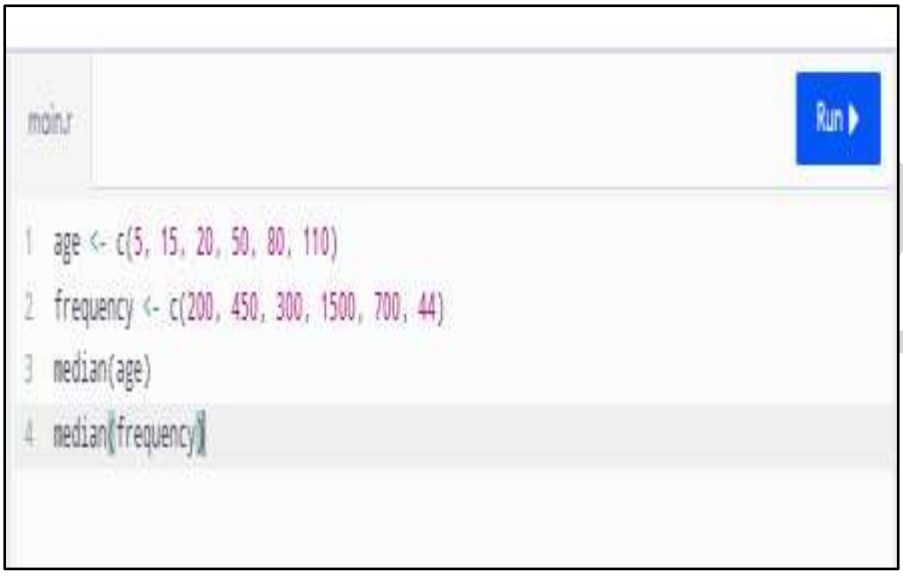
Date: 06/10/2026

Experiment-1

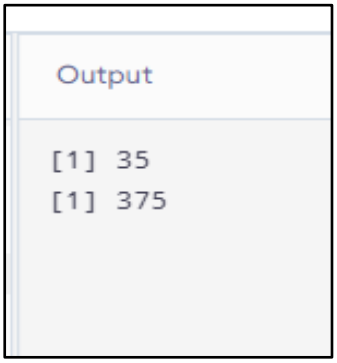
Aim:

To find intervals and corresponding frequencies.

Code:



Output:



Result:

It is executed successfully.

Experiment-2

Aim:

To find the data for analysis includes the attribute age.

Code:

main.r

Run

```
age<-c(13,15,16,16,19,20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70)
mean(age)
median(age)
mode_age<-names(table(age))[table(age)==max(table(age))]
mode_age
range(age)
quantile(age,.25)
quantile(age,.75)
```

Output:

192421292

Output

```
[1] 29.96296
[1] 25
[1] "25" "35"
[1] 13 70
 25%
20.5
75%
 35
```

Result:

It is executed successfully.

Experiment-3

Aim:

To show Data Preprocessing in Reduction and Transformation.

Code:

main.r

Run ▶

```r
data <- c(11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,30,40,45,45,45,71,72,73,75)
bins <- 5

bin_indices <- cut(data, bins)

mean_smooth <- tapply(data, bin_indices, mean)
print(mean_smooth)

median_smooth <- tapply(data, bin_indices, median)
median_smooth

min_max_smooth <- tapply(data, bin_indices, function(x) c(min(x), max(x)))
print(min_max_smooth)
```

Output:

Output

```
(10.9,23.8] (23.8,36.6] (36.6,49.4] (49.4,62.2] (62.2,75.1]
   17.78571    27.00000    43.75000          NA    72.75000
(10.9,23.8] (23.8,36.6] (36.6,49.4] (49.4,62.2] (62.2,75.1]
       19.5        27.0        45.0          NA        72.5
$`(10.9,23.8]`
[1] 11 23

$`(23.8,36.6]`
[1] 24 30

$`(36.6,49.4]`
[1] 40 45

$`(49.4,62.2]`
NULL

$`(62.2,75.1]`
[1] 71 75
```

Result:

It is executed successfully.